Radar Human Activity Detection

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CONCEPT OF OPERATIONS

CONCEPT OF OPERATIONS
FOR
Human Activity Radar Detector

TEAM 4

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Project Leader Date

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T/A Date

Change Record

Rev.	Date	Originator	Approvals	Description
-	9/17/2025	Radar Human Activity Detector		Draft Release

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1. Executive Summary

Tracking the activities of an individual in an assisted-living home is essential in ensuring their continued physical well-being. However, doing so often requires means that unintentionally violate the privacy rights of these individuals, such as the installation of equipment that collects visual, audio, and other forms of data. These are usually installed in private areas in which the individual cannot receive assistance in a timely manner. As such, it is critical that a product be developed that both reliably detects & alerts the proper personnel for emergency assistance and does not intrude on individual privacy. This product is designed to collect data using radar signal reflections to assess the received signal. Using a machine learning model trained on collected data sets of falling individuals, the fall signal will be continuously classified over a period of ten to twenty seconds. If the pattern persists, an alert will be transmitted to a mobile application interface for assisted-living personnel so that further emergency action can be taken. The scope of this project is to provide a prototype that increases the efficiency of response times by assisted-care workers at a minimal level of personal privacy violation.

2. Introduction

This document introduces the human activity radar detector, which monitors private areas in assisted-living homes for signs of fall injury in order to alert on-site personnel for further emergency care. The module will reduce the frequency and time between injury and the arrival of assistance and will ensure that the privacy of vulnerable individuals in these spaces is not violated.

2.1. Background

Injury from falling is common among the elderly in assisted-living homes, and such injuries can cause further physical ailments if the fall goes unnoticed. According to a collection of 18 studies on elderly individuals living in nursing homes, the incidence of falls is 43% [1]. This means that a significant minority of the elderly in these facilities have experienced some form of injury from falling. Therefore, it is critical to develop a product that minimizes the potential physical impairments that result from these injuries.

The protection of individual privacy is another key aim of this project. In recent times, the usage of camera systems for home security has led to privacy failures and breaches of trust on the part of service providers. For example, in 2023, the home security company Ring did not invest in adequate privacy & security provisions of customer data, which led to the misuse of the data by employees [2]. This presents a serious risk for users of home cameras, especially for those in assisted care who are more vulnerable, and a need for an alternative that can accomplish the same tasks without violating individual privacy. This project aims to develop a system that utilizes minimally invasive data collection, which will ensure the security of users' privacy.

2.2. Overview

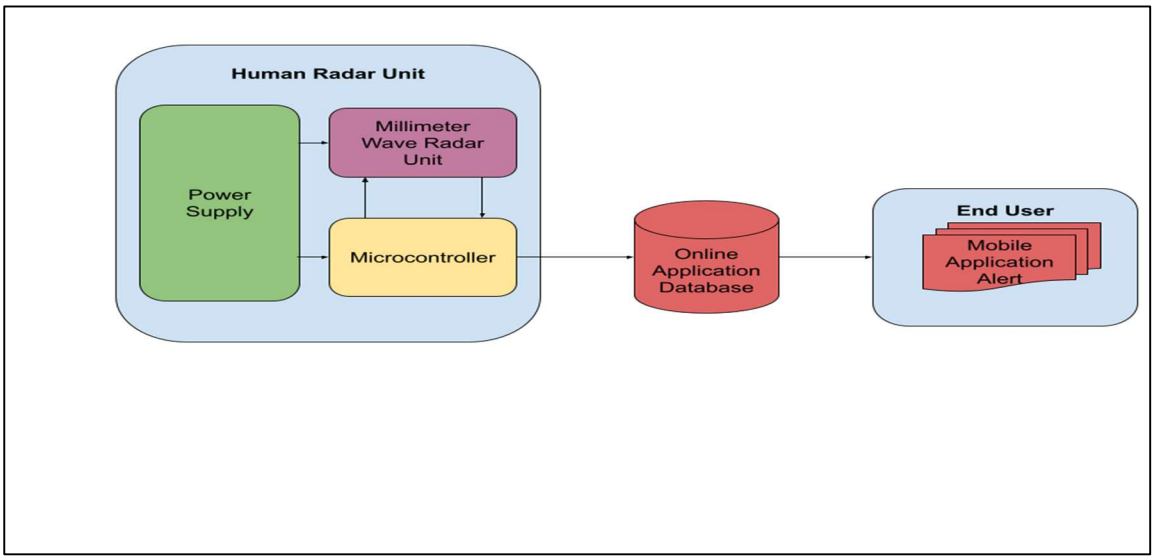

Figure 1: Human Activity Radar Detector

This system will be used to monitor a set location for people falling. This will be done by three primary components within the Human Radar Unit: the power supply, the microcontroller, and a mobile app interface to display the location of the received signal. The human activity radar detector will be outfitted with a radar module, which includes transmit and receive antennas, to detect objects at a range of eight to ten meters. This radar module will continually transmit & receive signals, which, using an onboard signal processing module, will send digital data to a microcontroller. The microcontroller will be programmed with a machine learning algorithm that will be trained on data sets of fall injury radar patterns. By classifying the received signal, the module will alert the on-site personnel through a mobile app interface so that they can provide the necessary care.

2.3. Referenced Documents and Standards

[1] Shao L, Shi Y, Xie XY, Wang Z, Wang ZA, Zhang JE. Incidence and Risk Factors of Falls Among Older People in Nursing Homes: Systematic Review and Meta-Analysis. J Am Med Dir Assoc. 2023 Nov;24(11):1708-1717. doi: 10.1016/j.jamda.2023.06.002. Epub 2023 Jul 8. PMID: 37433427.

[2] Staff, BCP, et al. "Ring's Privacy Failures Led to Spying and Harassment through Home Security Cameras." *Consumer Advice*, 30 Oct. 2023, consumer.ftc.gov/consumer-alerts/2023/05/rings-privacy-failures-led-spying-harassment-through-home-security-cameras.

3. Operating Concept

3.1. Scope

The Human Activity Radar Detector (HARD) will be used in residential and assisted living settings to detect human falls within a short distance (below 10m) using a high-frequency radar sensor. The radar sensor mounted on the wall will constantly be sending data to a control unit, where a microcontroller running a classification machine learning model will determine if there has been a fall injury. Once a fall injury has been detected, the microcontroller will communicate with an app via Wi-Fi to send out an alert to caretakers/family and emergency services.

3.2. Operational Description and Constraints

The Human Activity Radar Detector will be installed in spaces where assisted care is at the lowest level, such as bathrooms, bedrooms, etc. The module will transmit & receive signal reflections that will be processed to determine the occurrence of a falling injury, verify the certainty that the injury has in fact occurred, and alert the nearby assisted-care worker through an app interface. Below are listed the constraints of this design:

- The app interface must be able to identify the designated room location of a signal transmission in case of emergency.
- The module must be able to operate over the Wi-Fi network or otherwise be able to access an internet connection.
- The machine learning algorithm for detecting signal patterns will be trained on a predetermined set of data, which cannot be modified in the future.
- The system must be able to operate using a 9V battery that will be regulated by on-board circuitry.
- The system must continually classify affirmative signals for a period of 10-20 seconds before alerting personnel to prevent false alarms from triggering.

3.3. System Description

The radar detection system will be defined by three central parts: the radar detector, the microcontroller programmed with a machine learning algorithm, and the mobile application user interface. Power will be provided by a peripheral power management system. The radar detector will deliver digitally processed FMCW data to the microcontroller, which will classify the signal using a machine learning algorithm. The algorithm will decide whether the signal resembles a fall injury. If so, a signal will be transmitted over Wi-Fi to the user interface to indicate that a fall has occurred.

3.4. Modes of Operations

The system will have one primary mode of operation, in which detected falls are tracked with an alert sent to a mobile device. This will consist of three primary actions performed by the system: detection, validation, and finally alerting the user.

Detection
The system will emit & process signals continuously to identify whether an injury pattern has occurred.

Validation
The system will continue to process signals as before; however, the system will also maintain a counter that will verify that an injured person has remained on the floor after the fall.

Alert
The system will send an alert to the user interface to inform assisted-care personnel in the area that an injury has occurred.

3.5. Users

This system will primarily be used by individuals who are at risk of serious injury from falls and their immediate family or caretakers. The level of training required to use this system will be very low, with the only requirements being the installation of a mobile app and the mounting of the system on a wall. This system will benefit both at-risk individuals and their families, providing both of those groups the peace of mind that if something bad were to happen, they would be informed of it in a timely manner along with the appropriate medical personnel.

3.6. Support

This system will include a user manual in the box detailing how to install the physical device and the phone app. If users are having issues with the app, they will be able to either call a support number with their issue or send an email relaying the issue to the manufacturer.

4. Scenario(s)

4.1. Assisted Living Fall Detection

The Human Activity Radar Detector's primary use is for fall detection in assisted living environments. Residents in assisted living are at high risk of falls, which, if left unattended, can result in compounded side effects. The proposed radar system would provide continuous, non-invasive monitoring of desired areas in residential care homes (possibly in bedrooms/bathrooms where staff are less present). When falls are detected, the system will alert the staff via a phone/web app, ensuring a rapid response.

4.2. Gym Fall Detection

Another possible application of this system is to work within public gyms. Given the irregular hours that gyms can be open, having personnel on site constantly monitoring issues could be difficult. Areas or exercises where fall detection could be applicable include the squat rack, deadlift, power clean, and other such exercises with a full range of motion. Having an automated alert system could help employees watch for possible life-threatening issues while reducing mental workload

4.3. Child/Animal in Car Detection

Another application of radar detection could be detecting children or animals inside a locked car. There have been numerous cases globally of child or animal deaths due to being stuck in locked cars on a hot day. This issue is prevalent enough that, in the coming years, the EU will require cars to have radar detection systems inside for this exact purpose. There is no practical way for any police body to monitor all cars to protect against this issue. Having a system to check for this can ensure more child and animal lives are saved each year.

5. Analysis

5.1. Summary of Proposed Improvements

- The system will operate off battery power for its main configuration; this will allow placement in locations where a power source is not available.
- Ease of installation: Given that the system's logic is based on a machine learning model, user input will be minimal during both installation and operation.
- Cybersecurity integration: In order to prevent wireless communication that may retrieve user data, integrating basic cybersecurity into the system would greatly advance the stated purpose of protecting users' privacy.

5.2. Disadvantages and Limitations

- Need to ensure that batteries are adequately charged.
- Low computing capacity within the microcontroller limits ML implementation.
- Limited directionality compared to a camera-operated system.
- High cost for FMCW radar sensors.

5.3. Alternatives

Alternatives to the millimeter wave radar detection solution include, but are not limited to:

- Constant monitoring by attending caretakers

- Off-the-shelf camera monitoring systems
- The same unit, but with a wired power solution
- Life-Alert style button
- Smart watch fall detection alerts

5.4. Impact

This design is largely focused on reducing the impact of data collection on personal privacy; however, there remains a minimal level of data that must be collected from the operating environment in order for the design to function. This does not pose any significant risk to the user's personal information or privacy. In addition, the programmed algorithm does not feature the ability to receive updates, modifications, or any devices that record data that are not explicitly stated in this document, making the user's data secure from malpractice.

Human Activity Radar Detector

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INTERFACE CONTROL DOCUMENT

REVISION --
4 October 2025

INTERFACE CONTROL DOCUMENT
FOR
Human Activity Radar Detector

PREPARED BY:

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Author Date

APPROVED BY:

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Project Leader Date

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John Lusher II, P.E. Date

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T/A Date

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1	10/4/2025	Human Activity Radar Detector Team		Draft Release
2	12/06/2025	Human Activity Radar Detector Team		Raspberry Pi Added

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1. Overview

The Interface Control Document (ICD) for the Human Activity Radar Detector will provide more information on how the subsystems in the concept of operations and the functional system requirements (FSR) will be produced. It will also include more information on how each subsystem will interact with the others.

2. References and Definitions

Provide any references (i.e., standards documents) and definitions. Examples are shown below.

2.1. References

2.2. Definitions

PCB	Printed Circuit Board
mA	Milliamp
mW	Milliwatt
GHz	Gigahertz (1,000,000,000 Hz)
TBD	To Be Determined
UART	Universal Asynchronous Receiver-Transmitter
ML	Machine learning

3. Physical Interface

3.1. Weight

Component	Weight (kg)
Raspberry Pi 5	0.046
External Rechargeable Battery Pack	TBD
Radome	TBD
Radar Chip Board	0.119

Table 1*: System Component Weights*

3.2. Dimensions

	Length (mm)	Width (mm)	Height (mm)
Raspberry Pi 5	85.6	56.5	21

Table 2*: Microprocessor Dimensions*

3.2.1. Dimension of ML (MCU) subsystem

	Length (mm)	Width (mm)	Height (mm)
TI IWR1642	76.2	61.0	15.2

Table 3*: Radar Board Dimensions*

3.2.2. Dimension of Radar Board subsystem

	Length (in)	Width (in)	Height (in)
Rechargeable Battery Pack	TBD	TBD	TBD
Buck Converter System	TBD	TBD	TBD

Table 4*: Power Supply Dimensions*

3.2.3. Dimension of Power Supply subsystem

3.3. Mounting Locations

The device may be mounted in any location on a wall that is not exposed to electrical hazards, such as water. For placement in areas of operation with unavoidable contact with water, such as a bathroom, the device should be at a high altitude to prevent component damage.

4. Thermal Interface

Our device is expected to use a small heat sink for the purpose of cooling the microcontroller, which is the main heat-generating component of the system. All other components that generate a non-trivial amount of heat, such as the radar board, are expected to be adequately cooled by natural air cooling.

5. Electrical Interface

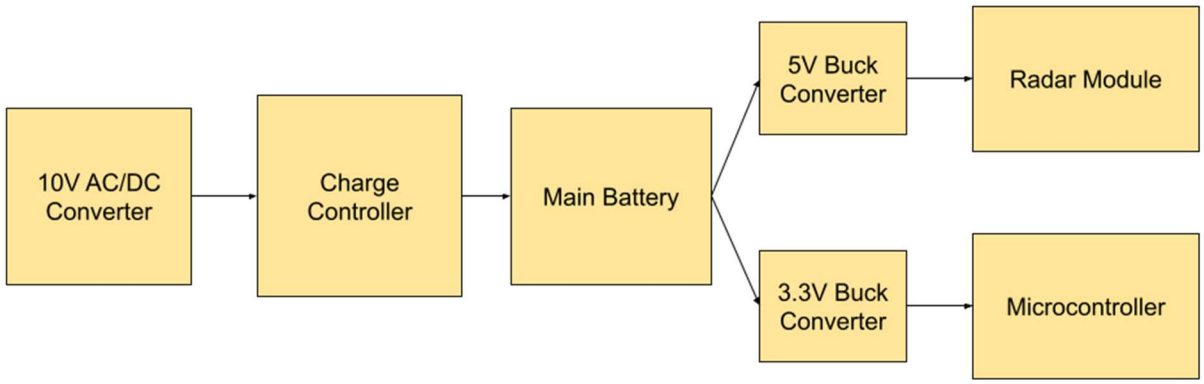

Figure 1: Power Distribution System

5.1. Primary Input Power

The primary power source for the system will be an internal rechargeable lithium-ion battery pack with a total voltage of 7.2V with a 7200mAh capacity. This battery will be charged using an AC/DC 10V 100W adapter which can be plugged into the device. As both the radar module and microcontroller have high power requirements, the device is expected to be able to run off battery power for at least 10 hours before needing to charge.

5.2. Voltage and Current Levels

Component	Voltage	Current (Max)	Power (Max)
Charge Controller (Charging Output)	7.2V	10A	72W
Radar Unit	5V	5A	25W
Raspberry Pi 5	5.1V	5A	25.5W

Table 5: Device Power Consumption

5.3. Signal Interfaces

Our system does not provide the user with a way to interact with the signals collected by the module to ensure the simplicity of design.

5.4. User Control Interface

The user will be able to interact with the system through the mobile/web application interface, where the individual modules will be instantiated and the modules can inform the user in the case of a detection event.

6. Communications / Device Interface Protocols

6.1. Wireless Communications (WiFi)

The microprocessor includes a built-in Wi-Fi system with integrated 2.4 GHz, 802.11 b/g/n Wi-Fi. Through this, the output determined by the ML model will be published to a cloud service that the application is subscribed to.

6.2. Device Peripheral Interface

The TI IWR1642 radar chip communicates with and sends data to the Raspberry Pi 5 through a serial port using UART.

Human Activity Radar Detector

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FUNCTIONAL SYSTEM REQUIREMENTS

REVISION --
4 October 2025

FUNCTIONAL SYSTEM REQUIREMENTS
FOR
Human Activity Radar Detector

PREPARED BY:

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Author Date

APPROVED BY:

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Project Leader Date

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John Lusher, P.E. Date

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T/A Date

Change Record

Rev.	Date	Originator	Approvals	Description
1	10/4/2025	Human Activity Radar Detector Team		Draft Release
2	12/06/2025	Human Activity Radar Detector Team		Raspberry Pi Added

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1. Introduction

1.1. Purpose and Scope

This document will prescribe the system requirements of this design in real-world operation. The purpose of the Human Activity Radar Detector (HARD) is to improve the efficiency of response times & privacy protection in assisted-care homes in the event of a fall injury. The system will improve response times by actively identifying a fall injury event using machine learning. By using a trained machine learning model, the system will classify fall injuries at a highly reliable rate. The system will provide the user with information about the location of detection in the event of injury. The system will collect the minimal amount of data necessary to ensure efficient detection. The system should ensure up to ten hours of typical operation before a battery recharge is necessary.

1.2. Responsibility and Change Authority

The functional requirements of the system should only be modified with the consensus of the team members. The details of implementation & methodology of subsystems may be modified upon the agreement of the team members. The team leader, William Jachimiak, will be responsible for making sure that the functional requirements of the system are met. The people responsible for each subsystem are listed below:

- William Jachimiak, PCB design and Microprocessor
- Jean Jacques Chirat, Machine learning algorithm & training
- Aiden George, Power system & distribution
- Preston Bowyer, User-end application interface

2. Applicable and Reference Documents

2.1. Applicable Documents

The following documents, of the exact issue and revision shown, form a part of this specification to the extent specified herein:

Document Number	Revision/Release Date	Document Title
47-CFR-95-M	9/20/2017	Permitting Radar Services in the 76-81 GHz Band
45-CFR-160,164	12/28/2000	Standards for Privacy of Individually Identifiable Health Information
ISO/IEC 29100:2024	02/2024	Security Techniques - Privacy Framework

2.2. Reference Documents

The following documents are reference documents utilized in the development of this specification. These documents do not form a part of this specification and are not controlled by their reference herein.

Document Number	Revision/Release Date	Document Title

2.3. Order of Precedence

In the event of a conflict between the text of this specification and an applicable document cited herein, the text of this specification takes precedence without any exceptions.

All specifications, standards, exhibits, drawings or other documents that are invoked as "applicable" in this specification are incorporated as cited. All documents that are referred to within an applicable report are considered to be for guidance and information only, except ICDs that have their relevant documents considered to be incorporated as cited.

3. Requirements

This section defines the minimum requirements that the development item(s) must meet. The requirements and constraints that apply to performance, design, interoperability, reliability, etc., of the system, are covered.

3.1. System Definition

The Human Activity Radar Detector is a fall detection system that will be able to detect a fall injury and send a notification via a web app to caretakers and family. It will be mounted to the wall in residential areas such as bedrooms and bathrooms. The four subsystems for the Human Activity Radar Detector are the power supply, microcontroller and PCB, the machine learning algorithm, and the web application.

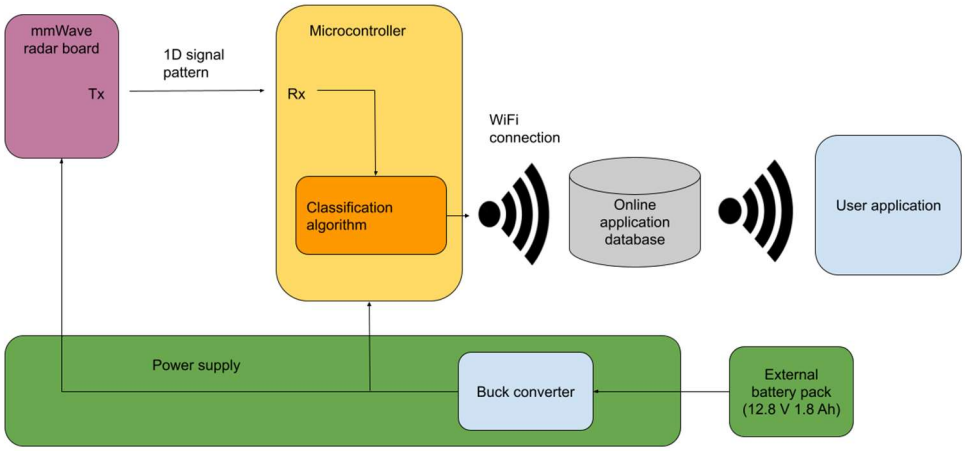

Figure 2: Block Diagram of System

The mmWave radar will transmit a signal consisting of the height component of a moving person and a range-doppler velocity component. The board will transmit this data to the Raspberry Pi 5 microprocessor, where the data will be processed through a simple neural network, with two criteria: a decrease in vertical coordinates and a downward velocity above an observed threshold. If it is decided that a fall injury occurred, the microcontroller will communicate with an online database, where the status of injury will be updated and the location of the particular module will be displayed for personnel to assist the injured.

3.2. Characteristics

3.2.1. Functional / Performance Requirements

3.2.1.1. Responsive in under 10 seconds

The system will be able to perform detection and alert the user within ten seconds of the occurrence of the fall injury event. The algorithm execution time, in addition to network latency, should remain under this limit in the vast majority of cases.

3.2.1.2. Low False Positive Rate

The system will have a low false positive rate of at most five to ten percent.

Rationale: A low false positive rate such as ~8% will prevent frequent disruption of other assisted-care responsibilities that on-site personnel may have.

3.2.1.3. Extremely low false negative rate

The system will have a very low to negligent false negative rate, and must be below 5% by requirement..

Rationale: The costs & liabilities of a false negative are far more consequential than a false positive. Therefore, it is imperative that false negatives are made as infrequent as possible for a reliable usage of the system.

3.2.1.4. Can run on battery for 10 hours

The Human Activity Radar Detector should be able to run on battery power and provide up to ten hours of active operation.

Rationale: The Human Activity Radar Detector will be entirely powered by battery and as a system that is designed to be able to detect at any time, it should be able to run for an extended period of time that is at least 10 hours.

3.2.2. Physical Characteristics

3.2.2.1. Installation

The installation of the radar detection module shall be specified in the user manual provided. The module will be pre-programmed with the trained algorithm.

3.2.2.2. Mass

The mass of the human activity radar detector shall be less than or equal to 4 kilograms.

Rationale: The system should be able to be hung on drywall using drywall screws with anchors that at minimum should be able to hold roughly 6 kg. Our limit for mass is less than this amount to ensure that there is minimal risk of damaging the equipment.

3.2.2.3. Volume Envelope

The volume envelope of the Human Activity Radar Detector shall be less than or equal to 7 inches in height, 6 inches in width, and 3 inches in length.

Rationale: This is a requirement specified by our customer due to constraints of their system in which the Search and Rescue System is integrating.

3.2.2.4. Mounting

The mounting information for the Search and Rescue System shall be captured in the Search and Rescue System ICD.

Rationale: As the Search and Rescue System mounts to platform system, the interface between the two includes mechanical, electrical and thermal details.

Electrical Characteristics

3.2.2.5. Inputs

The presence or absence of any combination of the input signals in accordance with ICD specifications applied in any sequence shall not damage the Human Activity Radar Detector, reduce its life expectancy, or cause any malfunction, either when the unit is powered or when it is not.

Rationale: By design, should limit the chance of damage or malfunction by user/technician error.

3.2.2.5.1 Power Consumption

The maximum peak power of the system shall not exceed 3W.

Rationale: This is to ensure that the module is not damaged by excessive voltage.

3.2.2.5.2 Input Voltage Level

The input voltage of the system will be 7.2V.

Rationale: Our radar board and microprocessor are designed to operate using 5V. This voltage level will be obtained using a stepdown buck converter before being supplied to the component.

3.2.2.6. Outputs

3.2.2.6.1 Data Output

The Human Activity Radar Detector will provide a mobile application for users to view detection alerts.

Rationale: The detection information passes directly to the caretaker's system.

3.2.2.6.2 Diagnostic Output

The Human Activity Radar Detector shall include identification for each radar module as well as the location of the fall injury.

Rationale: Eliminates the necessity of searching all module locations and pinpoints the exact location to facilitate a speedy response to the injured person.

3.2.2.6.3 Connectors

The Human Activity Radar Detector shall follow the American National Standard for Electrical Connectors ANSI C119.6-2011.

Rationale: Conform to connector standard.

3.2.2.6.4 Wiring

The Human Activity Radar Detector shall follow the guidelines set forth in the National Electric Code regarding electrical wiring.

Rationale: Conform to wiring standard.

3.2.3. Environmental Requirements

The Human Activity Radar Detector will operate within the minimal requirements for the environment specified in the section below.

Rationale: The HARD system will operate indoors where exposure to the elements will be minimal.

3.2.3.1. Thermal

The Human Activity Radar Detector will be able to operate in temperatures ranging from -20°C to 60°C. The module is expected to be stored in a room temperature environment, away from long-term exposure to outside conditions.

Rationale: The HARD system PCB may be damaged, or its operation hindered by exposure to abnormal temperature.

3.2.3.2. Humidity

The Human Activity Radar Detector will be able to operate in humidity ranging from 0% to 100%. The module is expected to be stored in a dry environment.

Rationale: The HARD system PCB may oxidize in a humid environment, which may seriously damage performance expectations.

3.2.4. Failure Propagation

The Human Activity Radar Detector shall not allow propagation of faults beyond the Human Activity Radar Detector user interface.

3.2.4.1. Failure Detection, Isolation, and Recovery (FDIR)

3.2.4.1.1 Built In Test (BIT)

The Human Activity Radar Detector shall be able to evaluate detection patterns and determine whether there is a failure.

3.2.4.1.1.1 BIT False Alarms

The Human Activity Radar Detector shall have a false alarm rate of less than 5 percent.

> *Rationale: This is a useful requirement for the purpose of limiting time spent by personnel reacting to false alarms.*

4. Support Requirements

The Human Activity Radar Detector requires a Wi-Fi connection and a functioning cloud database to operate, as well as a rechargeable battery with sufficient voltage level. The MCU will operate on the Wi-Fi network to transmit detection status to the cloud database.

Appendix A: Acronyms and Abbreviations

Below is a list of common acronyms and abbreviations, update based upon your project….

BIT	Built-In Test
FOV	Field of View
FSR	Functional System Requirements
GUI	Graphical User Interface
MCU	Main Controller Unit
Hz	Hertz
ICD	Interface Control Document
kHz	Kilohertz (1,000 Hz)
mA	Milliamp
Ah	Ampere-hour
GHz	Gigahertz (1,000,000,000 Hz)
W	Watt
PCB	Printed Circuit Board
TBD	To Be Determined
UART	Universal Asynchronous Receiver-Transmitter

Human Activity Radar Detector

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SCHEDULE

REVISION --

2 December 2025

Execution Plan for HARD Human Activity Radar Detector

	9-Sep	16-Sep	23-Sep	30-Sep	7-Oct	14-Oct	21-Oct	28-Oct	4-Nov	11-Nov	18-Nov	25-Nov	2-Dec
Learn and Understand the Problem													
Research and Planning										Not Started			
Write ConOps										In Progress			
Choose & order radar board										Completed			
Midterm Presentation										Late			
Write FSR & ICD													
Order Parts													
App UI Started													
Receive Parts													
Fire base cloud and database up													
Develop data preprocessing code													
Have Training Data Complete													

Complete PCB Design													
Status Update Presentation													
Automatic FCW Updates Working													
Develop code for ML													
Interface MCU to Radar													
Interface MCU to WIFI													
Evaluate ML model using data													
Apply Power Supply to PCB													
Apply MCU on PCB													
Final Presentation													
Validate Radar-MCU-WIFI													
Final Presentation													
Demo													

Human Activity Radar Detector

William Jachimiak

Jean Jacques Chirat

Preston Bowyer

Aiden George

VALIDATION PLAN

REVISION --

2 December 2025

Validation Plan for HARD Human Activity Radar Detector

Task	Specification	Subsystem/Owner	Status
Peak Power	10 W	Power Supply/Aiden	Completed
Maximum Charger Current	6 A	Power Supply/Aiden	Completed
Maximum Battery Life	10 hr	Power Supply/Aiden	Completed
Correct Frames without Drops	≥ 99%	Microprocessor/William	Completed
Wi-Fi Output Signals Sent Out and Received	≥ 99%	Microprocessor/William	Completed
Radar Detect Distance	8m	ML/Jean Jacques	Completed
Detection False Positive Rate	≤ 5%	ML/Jean Jacques	Completed
Detection False Negative Rate	≤ 5%	ML/Jean Jacques	Completed
Detection Processing Delay	≤ 8 s	ML/Jean Jacques	Completed
Push Notification delivery success	≥ 99%	App UI/Preston	Completed
Multi-device fan-out	± 300ms spread	App UI/Preston	Completed
Notification time	~ 5 seconds	App UI/Preston	Completed

Human Activity Radar Detector

William Jachimiak

Jean Jacques Chirat

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Aiden George

SUBSYSTEM REPORTS

REVISION --

2 December 2025

SUBSYSTEM REPORT

FOR

Human Activity Radar Detector

PREPARED BY:

_____

Author Date

APPROVED BY:

_____

Project Leader Date

_____

John Lusher, P.E. Date

_____

T/A Date

Change Record

Rev.	Date	Originator	Approvals	Description
-	12/6/2025	Human Activity Radar Detector Team		Draft Release

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1 Introduction

Our project will monitor the physical safety of fall injury prone individuals in assisted care using minimally invasive methods. The system collects vertical position and velocity data using a TI mmWave radar. The radar streams the data packets to a Raspberry Pi 5, which will run a machine learning detection model, which is trained on collected datasets employing the same radar module and using the same type of data. The Raspberry Pi communicates with an online Firebase database over the Wi-Fi network in order to transmit the detection result. The detection result, if positive, is displayed by an Android app which has an interface with Firebase. The app will notify the caretaker responsible with an alert displaying that a fall has occurred along with the room location in the event of an injury.

2 Microprocessor Unit Subsystem Report

2.1 Subsystem Introduction

The microprocessor subsystem of the Human Activity Radar Detector acts as a means of interfacing the TI mmwave IWR1642BOOST radar board with the machine learning subsystem and then interfacing the results of that with the app subsystem. The interface between the radar chip and the microprocessor was validated through the computer console, showing both what is being transmitted by the microprocessor and what is being received and comparing that to expected results. The interface between the app and the microprocessor was validated using a webhook displaying the JSON contents.

2.2 Subsystem Details

The microprocessor uses UART to communicate with the TI mmwave radar board using the boards built in micro usb to UART port. The TI mmwave radar board uses The Raspberry Pi sends the configuration details to the TI radar board and once it receives back positive confirmation that the board is correctly configured, it takes data from the board at 20 FPS and processes the data. This is shown in Figures 1 and 2. From there, the microprocessor sends a JSON type file over Wi-Fi whenever it receives the push of a button (which simulates a positive fall detection from the machine learning subsystem) which is shown in Figure 3.

```
b'sensorStop\nIgnored: Sensor is already stopped\n\rDone\n\rmmwDemo:/>flushCfg\nDone\n\rmmwDemo:/>dfeDataOutputMode 1\nDone\n\r
mmwDemo:/>channelCfg 15 3 0\nDone\n\rmmwDemo:/>adcCfg 2 1\nDone\n\rmmwDemo:/>adcbufCfg -1 0 1 1 0\nDone\n\rmmwDemo:/>profileCfg
 0 77 79 7 18.24 0 0 82.237 1 64 6249 0 0 30\nDone\n\rmmwDemo:/>chirpCfg 0 0 0 0 0 0 0 1\nDone\n\rmmwDemo:/>chirpCfg 1 1 0 0 0
0 0 2\nDone\n\rmmwDemo:/>frameCfg 0 1 128 0 50 1 0\nDone\n\rmmwDemo:/>lowPower 0 1\nDone\n\rmmwDemo:/>guiMonitor -1 1 1 0 0 0 0
\nDone\n\rmmwDemo:/>cfarCfg -1 0 2 8 4 3 0 15 1\nDone\n\rmmwDemo:/>cfarCfg -1 1 0 8 4 4 1 15 1\nDone\n\rmmwDemo:/>multiObjBeamF
orming -1 1 0.5\nDone\n\rmmwDemo:/>clutterRemoval -1 0\nDone\n\rmmwDemo:/>calibDcRangeSig -1 0 -5 8 256\nDone\n\rmmwDemo:/>exte
ndedMaxVelocity -1 0\nDone\n\rmmwDemo:/>bpmCfg -1 0 0 1\nDone\n\rmmwDemo:/>lvdsStreamCfg -1 0 0 0\nDone\n\rmmwDemo:/>compRangeB
iasAndRxChanPhase 0.0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0\nDone\n\rmmwDemo:/>measureRangeBiasAndRxChanPhase 0 1.5 0.2\nDone\n\rmmwD
emo:/>CQRxSatMonitor 0 3 4 31 0\nDone\n\rmmwDemo:/>CQSigImgMonitor 0 31 4\nDone\n\rmmwDemo:/>analogMonitor 0 0\nDone\n\rmmwDemo
:/>aoaFovCfg -1 -90 90 -90 90\nDone\n\rmmwDemo:/>cfarFovCfg -1 0 0 9.11\nDone\n\rmmwDemo:/>cfarFovCfg -1 1 -5 5.00\nDone\n\rmmw
Demo:/>calibData 0 0 0\nDone\n\rmmwDemo:/>sensorStart\nDebug: Init Calibration Status = 0x17fe\n\rDone\n\rmmwDemo:/>'
```

Figure 3: Received Configuration Confirmation from the TI Radar Board

```
headerStartIndex    = 0
totalPacketNumBytes = 320
platform            = b'000a1642'
frameNumber         = 667
timeCpuCycles       = 895127384
numDetObj           = 5
numTlv              = 3
subFrameNumber      = 0
The 1st TLV
    type 1
    len 80 bytes
The 2nd TLV
    type 7
    len 20 bytes
                 x(m)         y(m)         z(m)        v(m/s)    Com0range(m)  azimuth(deg)  elevAngle(deg)  snr(0.1dB)    noise(0.1dB)
    obj  0:    -0.122466     1.955622     0.000000     0.000000     1.959453    -3.583322            0          155          881
    obj  1:     0.612329     1.861319     0.000000     0.000000     1.959453    18.209957            0          155          881
    obj  2:     0.022267     0.176735     0.000000     0.618046     0.178132     7.180756            0          440          665
    obj  3:    -0.050100     0.170942     0.000000     0.618046     0.178132   -16.334824            0          440          665
    obj  4:     2.961445     1.638519     0.000000    -0.077256     3.384509    61.044972            0          225          741
Detected objects: 5
obj0: x=-0.12246578931808472, y=1.9556218385696411, z=0.0, v=0.0, snr=155
obj1: x=0.6123289465904236, y=1.861318826675415, z=0.0, v=0.0, snr=155
obj2: x=0.022266507148742676, y=0.17673492431640625, z=0.0, v=0.6180459260940552, snr=440
obj3: x=-0.05009964108467102, y=0.17094166576862335, z=0.0, v=0.6180459260940552, snr=440
obj4: x=2.9614453315734863, y=1.6385185718536377, z=0.0, v=-0.0772557407617569, snr=225
```

Figure 4: An Output Frame showing Position, Velocity, and Number of Detected Objects

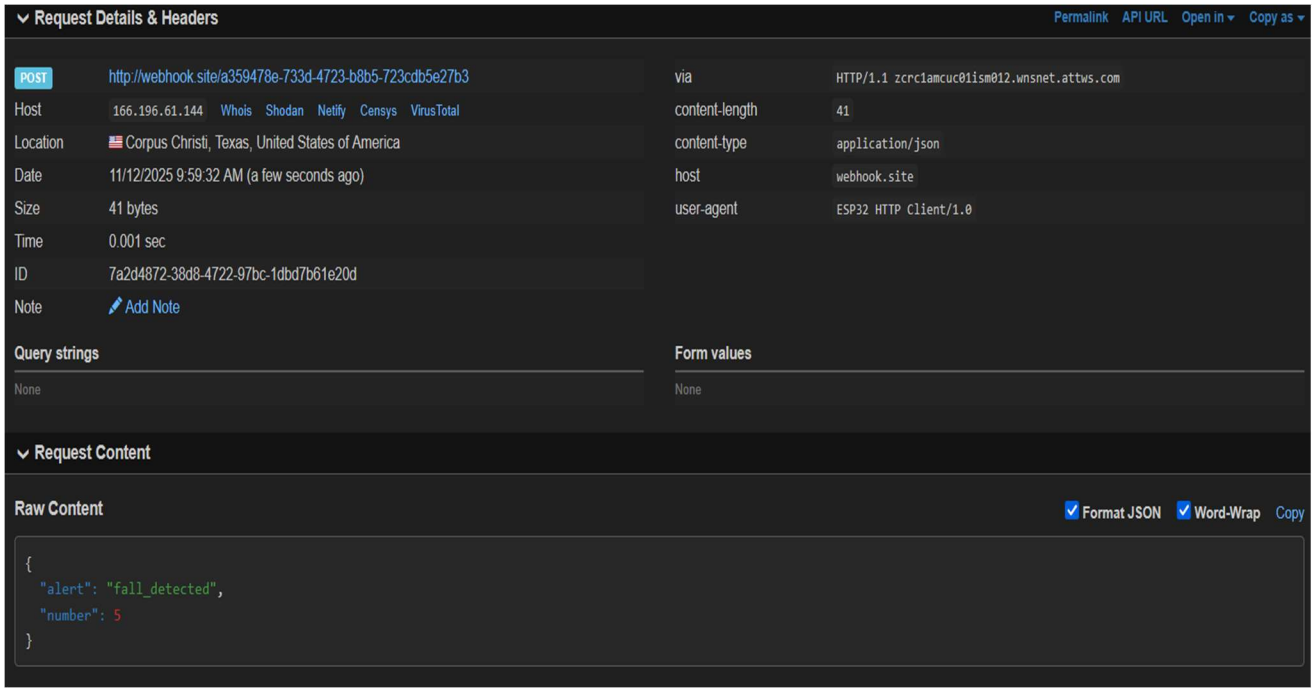

Figure 5: Webhook Showing the JSON file Contents

2.3 Subsystem Validation

The most important metrics to be validated for the microprocessor were to ensure that the data coming in from the TI radar board was taken in every frame at 20 FPS, and no frames were being dropped, along with frames being correctly defined. Figure 4 below shows an example of frames being dropped as a result of 0 objects being detected. Due to late complications with the initial design that utilized an ESP32-S3 microcontroller, the decision to switch to a Raspberry Pi 5 was made too late to properly validate subsystem using the Raspberry Pi. As such, the validation was done using an ESP32-S3 development board connected to the computer, and the computer was simultaneously connected with the TI radar board. In this setup, the computer acted as a middleman between the ESP32 and the radar board. Table 1 below details the validation plan.

Validation Criteria	Definition	Result
Correct Frames without Drops	>99% of Frames are Correctly Processed	Pass
Configuration Verified	Configuration Validated before Data is taken in	Pass
Wi-Fi Output Signals	>99% of JSON signals sent via Wi-Fi are correctly received	Pass

Table 1: Microprocessor Subsystem Validation Plan

```
************ Frame Fail, numDetObj = 0 ****************
headerStartIndex    = 0
totalPacketNumBytes = 192
platform            = b'000a1642'
frameNumber         = 19
timeCpuCycles       = 311082973
numDetObj           = 0
numTlv              = 2
subFrameNumber      = 0
************ Frame Fail, numDetObj = 0 ****************
headerStartIndex    = 0
totalPacketNumBytes = 192
platform            = b'000a1642'
frameNumber         = 19
timeCpuCycles       = 311082973
numDetObj           = 0
numTlv              = 2
subFrameNumber      = 0
```

Figure 6: Frame Failure

2.4 Subsystem Conclusion

The subsystem works as expected within the scope of the validation plan. The subsystem still needs to be fully moved over to the Raspberry Pi 5 before it can be fully integrated and run independently of an external computer. Beyond that, there is more work to be done in terms of matching the existing interfaces of each subsystem with each other.

3 Power Supply Subsystem Report

3.1 Subsystem Introduction

The power supply is designed to convert a DC voltage battery into two separate DC voltage rails. This battery is charged using a battery charging IC that converts an AC input into a DC output. The power supply was only tested in battery configuration due to delays in the delivery of IC components and general time restrictions.

3.2 Subsystem Details

The power supply subsystem consists of 3 main components, the battery charging IC, the battery, and the two step down converters for turning the 7.2V input power into 5V and 3.3V output rails respectively. The following graph:

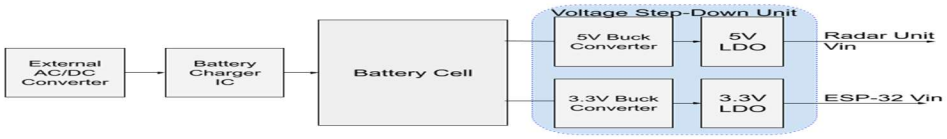

Figure 7: 5V and 3.3V Buck Converter Schematic

3.3 Subsystem Validation

To validate the system's output, the design was first tested using OrCad for its PSpice-for-Ti plug ins. When testing the design, several concessions had to be made, as the battery charger IC was too complex for simulation as stated by Texas Instruments documentation. Because of this, only the two step-down converters could be tested and validated as shown below with both the initial transient battery connect for the 5V and 3.3V rail shown below.

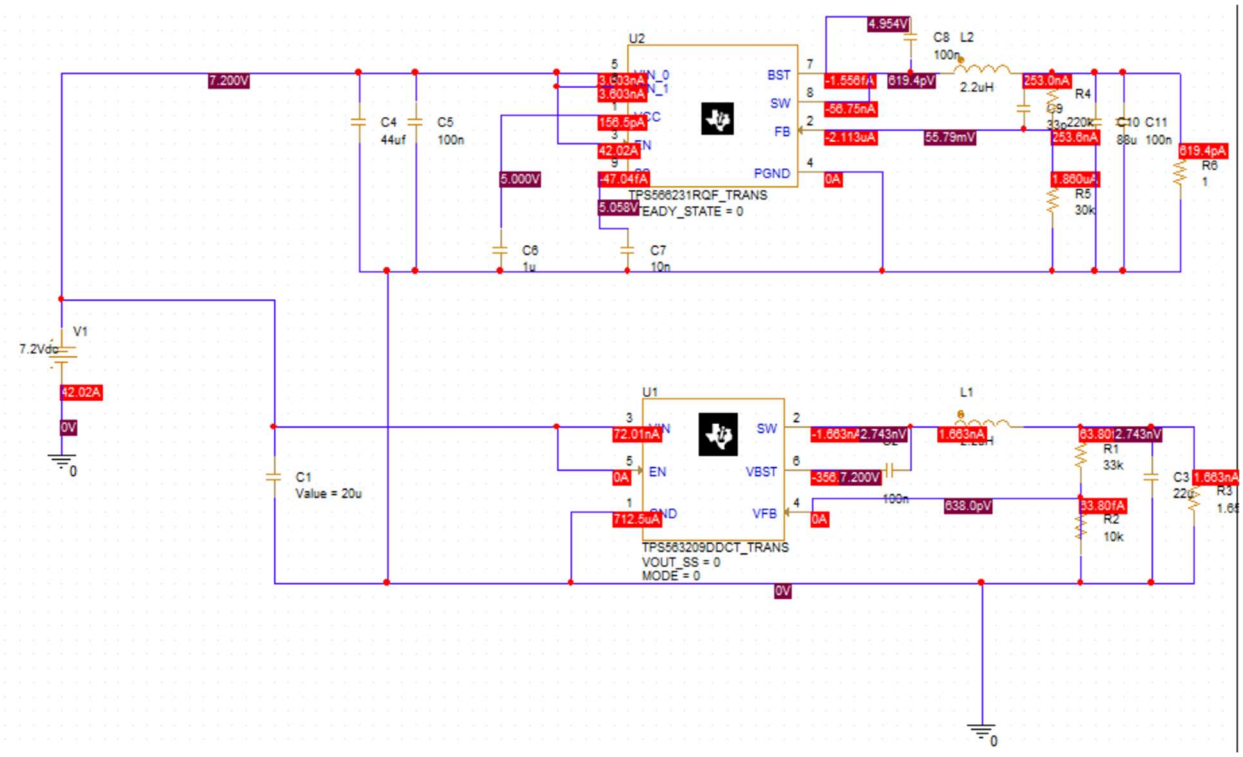

Figure 8: 5V and 3.3V Buck Converter Schematic

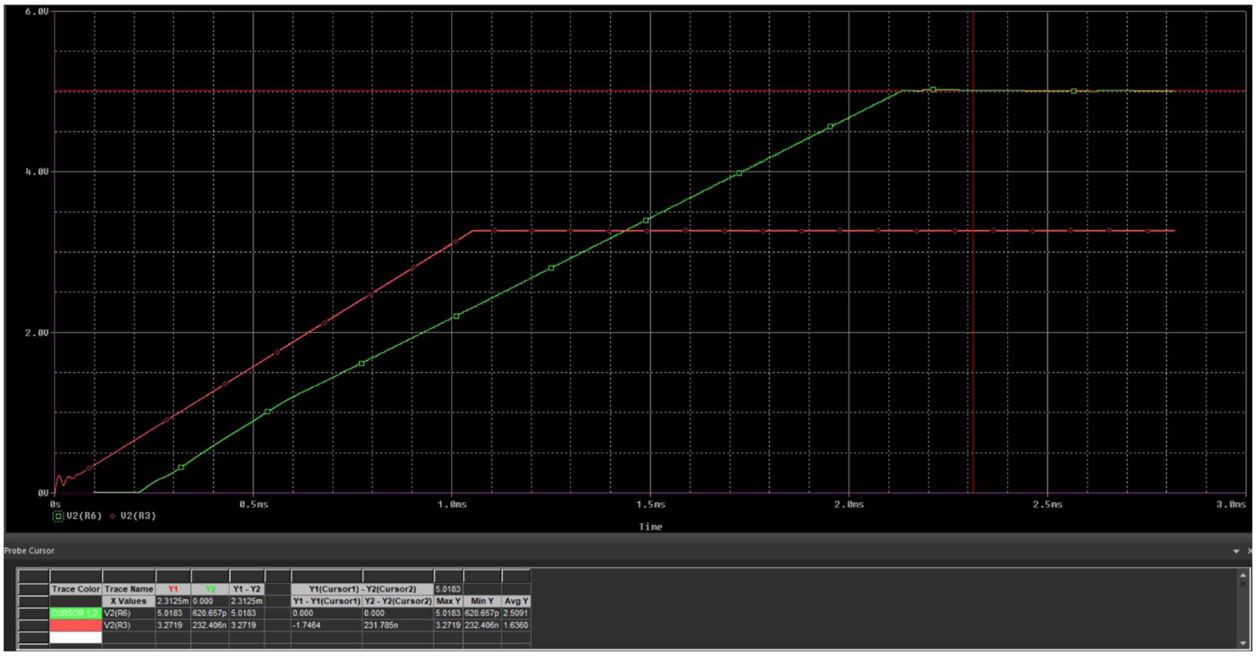

Figure 9: Buck Converter Voltage Simulation

Figure 10: Buck Converter Current Simulation

From this, it could be seen that functionality of the voltage converters was expected. From this, the 3.3V and 5V rails were tested at the expected voltage range of 5-8V for the 3.3V rail line and 6-8V for the 5V rail line both with a standard power supply and testing with the 7.2V 5000mA battery connected. When the battery was connected, a multimeter registered a 3.3V output and a 6.3V output. From this, it can be seen that the 5V rail is

outputting the incorrect voltage. Looking into Texas Instruments documentation for the TSP566238, the output voltage is equal to the following.

$$V_{out} = 0.6\left(1 + \frac{R_{Upper}}{R_{Lower}}\right)$$

Therefore, future modifications will require the replacement of the upper and lower resistors for further troubleshooting of the 5V line output.

Figure 11: Battery Connected Voltage Divider Output

3.4 Subsystem Conclusion

Testing successfully validated the peak current capabilities of both rails and the voltage of the 3.3V rail, but the 5V rail failed, outputting 6.3V due to a resistor divider error. Additionally, the battery charging IC was not tested due to issues in the voltage divider taking priority for testing. Finalizing the subsystem requires correcting the 5V rail and replacing the 3.3V rail with a secondary 5V rail to account for the new microcontroller.

4 Machine Learning Subsystem Report

4.1 Subsystem Introduction

The machine learning subsystem is designed to reliably carry out the core of the fall detection task. The model used consists of a convolutional neural network trained on data collected from the radar board. The datasets were collected with the intention of capturing a wide range of what possible elderly fall injuries could look like, which includes many edge cases. Using standard neural network methods such as a validation & testing dataset, set aside to progressively evaluate the performance of the model across training iterations, the detection capability of the model was verified to function in close accord with the specifications required by the system.

4.2 Subsystem Details

Below is a block diagram of the ML subsystem

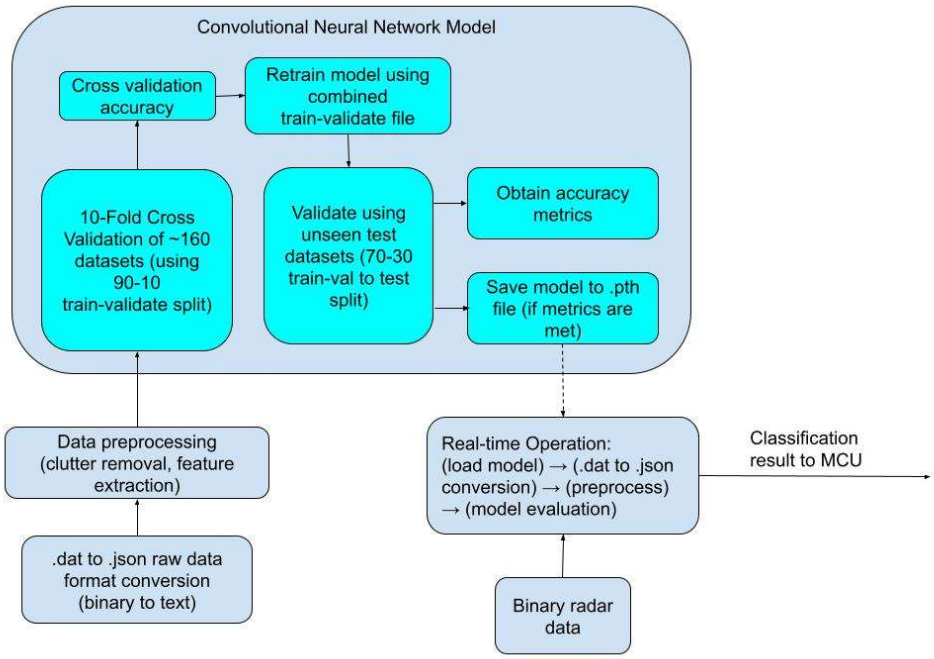

Figure 12: Machine Learning Subsystem Block Diagram

The design choices for the machine learning subsystem are as follows in the following sections. To preface, it was determined that a neural network would be an efficient architecture to perform the classification/detection task that the project requires, due to the high variance in what fall injuries could look like, rather than decision trees like random forest or linear regression methods.

4.2.1 Data Formatting

For fall detection, it was determined that tracking statistical features such as average vertical position, average doppler velocity, and downward acceleration of a radar cross section would be necessary for the model to distinguish between falls and non-falls. The data undergoes a process of clutter removal (for stationary objects and reflections) and moving averaging in order to consolidate a clear object for detection. In addition to this, the network utilizes the 256 range bin values provided by the radar's internal processing which is also featured in the model input. These values measure the signal strength in 256 increments of distance from the radar, and therefore prove useful in tracking the motion pattern of an individual.

4.2.2 Neural Network Architecture

A convolutional neural network using a 1D convolution method was chosen as the specific architecture as it is well suited for sequential data like the radar's output. The network operates by sliding a kernel size of reasonable proportions (~1 second of data), performing a convolution filter across all frames of data. The model is flattened using max pooling, and then processed through a typical hidden layer network. The output of the network is a binary value indicating whether a fall (1) or non-fall (0) has occurred for a particular batch of data. Below is an example of training vs. validation loss for a typical training process:

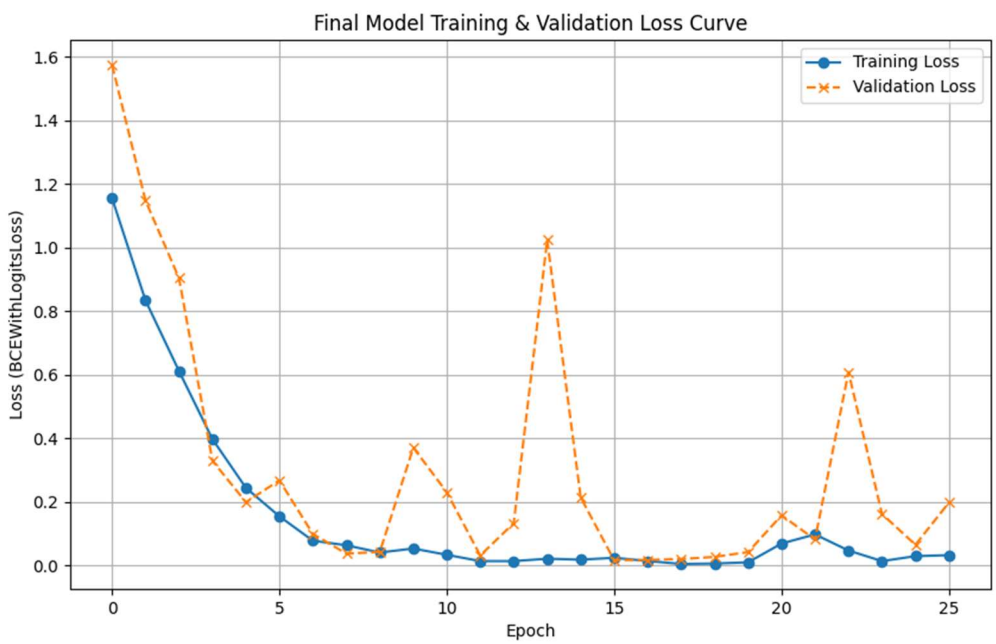

Figure 13: Training & Validation Loss Curves

4.2.3 K-Fold Cross Validation

A 10-fold cross validation method (which retrains the model across different permutations of training files) was implemented to ensure that the training data is sufficiently diverse for detection of edge cases. The method utilizes the combined train-validation set which comprises 70% of all recorded data and does a train-validation split of 90-10 for each fold. From here, the model is retrained 10 times, with the validation set moved around each time so that validation is applied to as many datasets as possible. Doing so allows the model parameters & output accuracy to be assessed as a whole, rather than an arbitrary split which may or may not produce desirable results, depending on what files end up where.

Below is an example plot of validation accuracy across 10 folds:

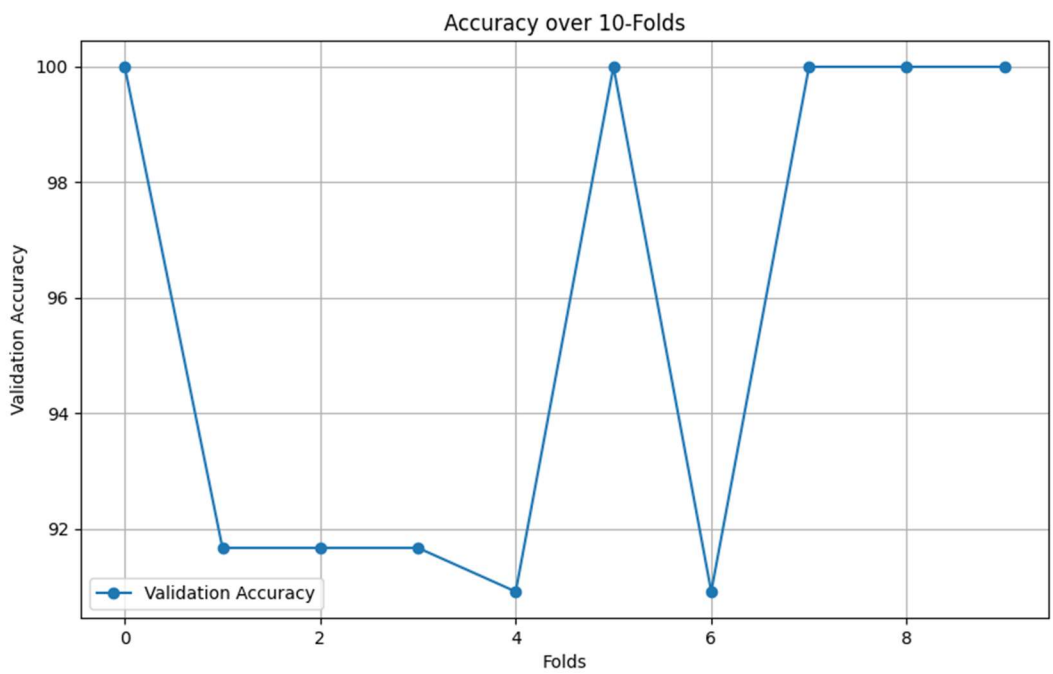

Figure 14: 10-Fold Cross Validation Accuracy

4.3 Subsystem Validation

The most important metrics for the subsystem are the false positive and false negative rates. The false positive rate must be small, since false positives cost resources. The false negative rate must also be small, since missing a fall could have fatal consequences for the user. The subsystem is validated using a test dataset, which consists of 30% of the total data. This split was chosen to ensure that enough edge

cases would be included in the test data, whilst also ensuring the model has enough training data to be robust.

Below are some performance criteria for the subsystem:

Validation Criteria	Definition	Progress (as of latest model)
False Positive Rate	A non-fall event which is classified as a fall event as a ratio of all non-fall cases	2.99%
False Negative Rate	A fall event which is classified as a non-fall event as a ratio of all fall cases	2.56%
Train & Test Dataset Size	The combined number of datasets used for model training and datasets for model simulation	164
Fall Positive Dataset	Number of datasets which serve as ground truth for falling scenario or '1'	60
Fall Negative Dataset	Number of datasets which serve as ground truth for non-falling scenario	104
Edge case handling	Model must account for motion patterns that do not conform to majority of fall or non-fall patterns	20
Confuser Rejection	Combined total of false positive and false negative classifications as a ratio of total test cases	2.83%
Noise handling	Model must account for noise that may be present in real-life residences	Added Gaussian noise of magnitude 0.1
Detection delay	Model must be able to completely process any falling motion data within a reliable timeframe	TBD in integration
Computation Speed	Real-time data must be converted to .json, preprocessed, and classified before the processing of the following data batch. A standard data batch will be 160 frames, so a dataset of this length will be used to set the benchmark.	0.313 s

Table 2: Machine Learning Subsystem Performance Criteria

A screenshot of terminal output to validate that the model can perform at the metrics specified above:

```
CROSS-VALIDATION SUMMARY

Training Accuracy: 100.00% ± 0.00%
Validation Accuracy: 97.09% ± 4.45%

Aggregated Confusion Matrix (all validation folds):
[[65  2]
 [ 1 38]]
True Negatives:  65 (61.3%)
False Positives: 2 (1.9%) - false alarms
False Negatives: 1 (0.9%) - MISSED FALLS
True Positives:  38 (35.8%)
Fall Recall:     97.4% (sensitivity)
Fall Precision:  95.0%

FINAL TEST SET EVALUATION

Retraining on full train+val set...
 Early stopping at epoch 26

Test Accuracy: 98.28%
Test Confusion Matrix:
[[37  0]
 [ 1 20]]
```

Figure 115: Terminal Output for Model Training Run

The false positive, false negative rates, and computation speed are within reasonable margins for a reliable operation of the detection system.

4.4 Subsystem Conclusion

The most critical components of the subsystem are shown to be working correctly. The neural network is able to learn from given datasets and classify new data at a high rate of accuracy and do so within a reasonable timeframe.

5 App & Interface Subsystem Report

5.1 Subsystem Introduction

The mobile application subsystem of the HARD (Human Activity Radar Detection) system provides the interface through which caregivers receive, and review fall-detection alerts generated by the radar hardware. Its primary purpose is to deliver real-time notifications, present relevant alert information, and record user acknowledgements to support future system analytics.

This subsystem is implemented as an Android application built with Flutter and backed by Firebase services, including Authentication, Cloud Firestore, and Firebase Cloud Messaging (FCM). The app organizes alerts into categorized tabs, supports secure user access, and enables direct navigation from push notifications to detailed alert views. In addition, it includes an analytics page that summarizes alert activity and response trends.

Together, these features ensure that the HARD system can communicate critical events quickly and reliably to end users while providing a scalable foundation for future integration and enhancements.

5.2 Subsystem Details

5.2.1 Mobile Application Infrastructure

The mobile application for the HARD system was developed using Flutter and deployed on Android devices. Flutter was chosen because it provides a fast development cycle and a clean, consistent interface across different screens and device types. Since the goal of this subsystem is to provide caregivers with real-time access to fall alerts, the app needed a backend that could handle quick data updates and reliable user authentication. For that reason, Firebase was selected as the supporting cloud platform.

The app uses Firebase Authentication for secure sign-in, Cloud Firestore to store and retrieve alert information, and Firebase Cloud Messaging (FCM) to deliver push notifications. These services work together so that as soon as the radar and embedded subsystem generate a fall event, the alert appears in the app almost immediately—either as a notification or directly in the alert list if the app is already open.

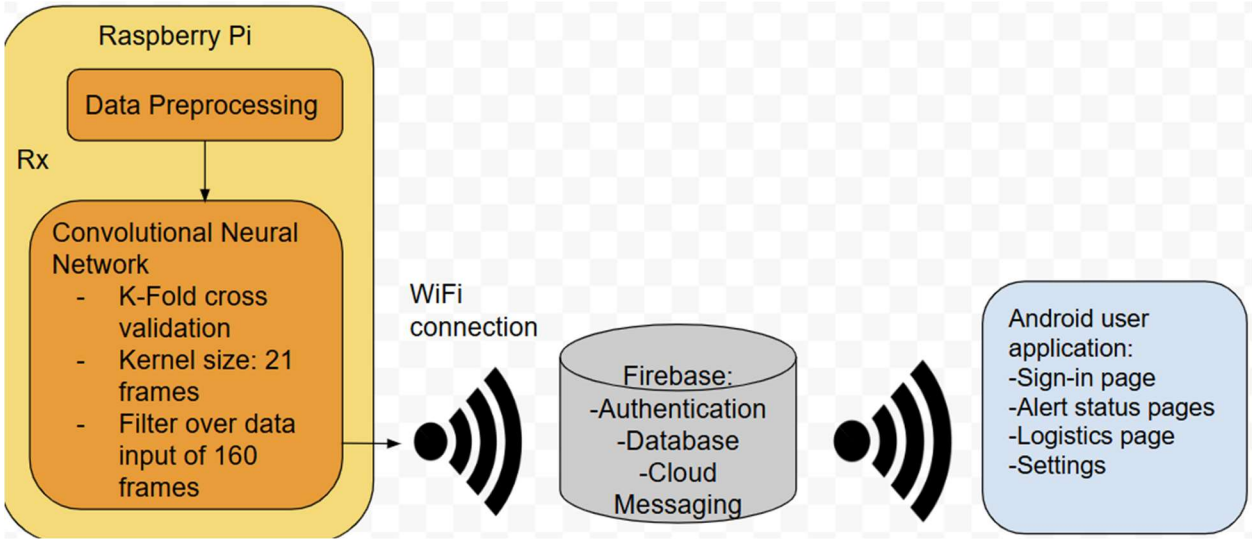

Figure 116: HARD Mobile Subsystem Data Flow

5.2.2 Data Model and Storage Design

All alerts presented in the app are stored in a single Firestore collection named **alerts**. Each alert is saved as its own document, which makes it easy to add or modify fields without needing to change any database structure. Every document includes the time the alert was generated, the type of event, optional fields such as room number or confidence score, and several acknowledgement-related fields used by the app.

Field	**Meaning**
timestamp	Time the alert was created
type	Type of detection event
room	Optional location information
confidence	Optional confidence value from the AI detection
acknowledged	Indicates whether the alert has been reviewed
acknowledgedBy	User who acknowledged the alert
acknowledgedAt	Timestamp of the acknowledgement

Table 3: App User Field Statistics

The app relies heavily on this structure to populate the Unacknowledged and Acknowledged tabs. Whenever a user acknowledges an alert from the detail screen, the app updates the Firestore document to reflect who responded and when. This approach keeps all data synchronized in real time and builds a foundation for later analytics throughout the project.

5.2.3 Software Architecture Stack

The software stack for the mobile subsystem combines Flutter's UI framework with several Firebase and third-party packages. Flutter handles all user interface elements, navigation between screens, and the overall styling of the app. Firebase's libraries provide the backend functionality—initializing the app, validating user credentials, reading and writing Firestore data, and receiving cloud messages.

Additional packages support specialized features. The flutter_local_notifications package allows the app to display Android notifications when a fall alert arrives while the app is open. The intl package is used to format timestamps into readable date/time strings. The fl_chart package powers the bar charts and summary graphics in the analytics tab.

Together, these components form a lightweight but capable architecture that can grow as the HARD project expands.

5.2.4 Application Interface and Functional Design

The mobile app is organized around three main tabs that separate alerts by their current status and provide additional context for system behavior.

Unacknowledged Alerts:
This tab displays all active fall alerts that have not yet been reviewed by a user. The information updates automatically as new data is written to Firestore. A small red badge on the tab icon indicates how many unacknowledged alerts are currently waiting.

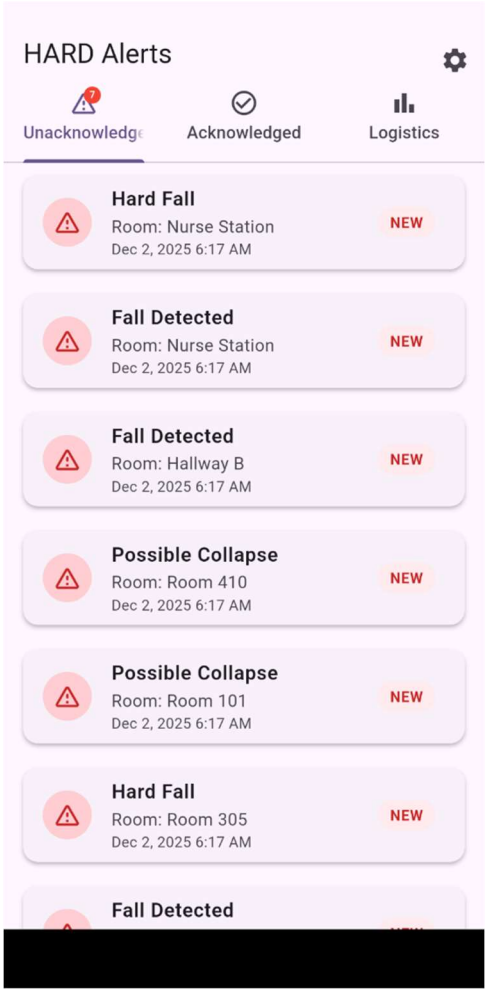

Figure 17: Unacknowledged Alerts Tab

Acknowledged Alerts:
Past alerts that have already been addressed are shown here. This gives users a simple history of events and allows quick verification that an alert was acknowledged correctly.

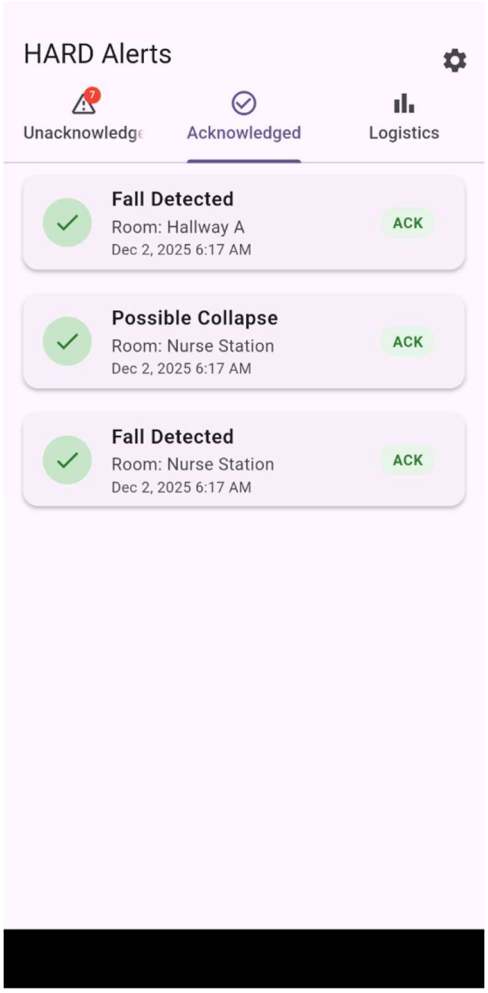

Figure 185: Acknowledged Alerts Tab

Analytics Tab:
This tab summarizes system activity using charts and room-by-room statistics. The app calculates totals, identifies the most active rooms, and estimates average response times by comparing alert timestamps to acknowledgement times. A bar chart provides a visual overview of alert frequency, and users can tap into individual rooms to see more detailed information.

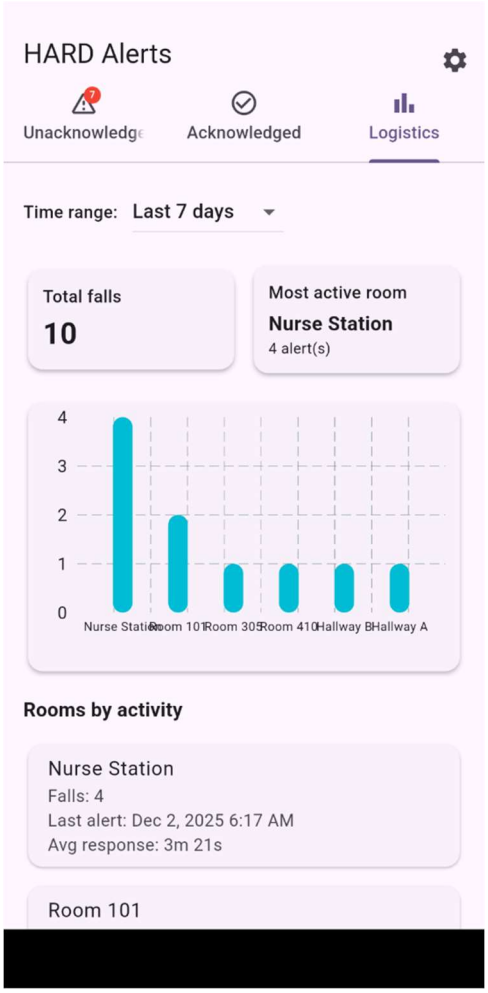

Figure 19: Logistics/Analytics Page

In addition to the three main tabs, the app includes a login page for user authentication, a detailed alert page that displays all information for a single alert, and a settings page accessible from the top-right corner of the home screen. The app also supports deep linking from notifications—when a user taps a notification, they are taken directly to the relevant alert detail screen.

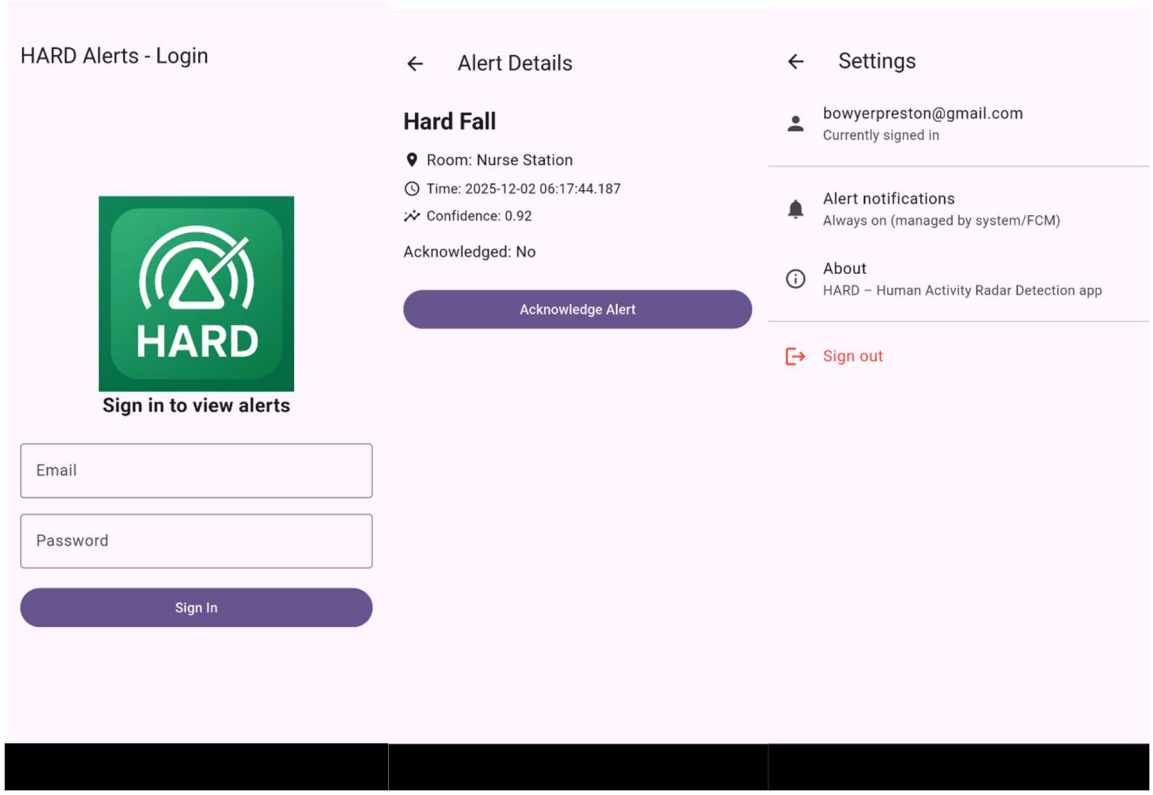

Figure 20: Login, Alert Details, and Settings Pages

5.3 Subsystem Validation

Validation of the mobile application subsystem focused on confirming that each core feature operated correctly and interacted consistently with Firebase. User authentication was tested by attempting both valid and invalid logins, which verified that Firebase Authentication handled credential checks and error conditions as expected. To validate data retrieval, sample alert documents were added to Firestore, and the application was observed to ensure they appeared immediately in the Unacknowledged tab, confirming that the real-time listeners and queries were functioning properly.

Push notification behavior was evaluated with the app in the foreground, background, and fully closed states. In all scenarios, Firebase Cloud Messaging successfully delivered notifications, and tapping them correctly routed the user to the alert's detail screen. Local notifications were also confirmed to work when the app was active. The acknowledgement workflow was validated by acknowledging test alerts and confirming that Firestore updated the acknowledged, acknowledgedBy, and acknowledgedAt fields, followed by the alert automatically transitioning to the Acknowledged tab.

Finally, the analytics tab was tested by generating alerts across multiple rooms and confirming that the subsystem calculated alert counts, last-alert timestamps, and response times accurately. The bar chart and room-level summaries updated to reflect these changes. These results demonstrated that the mobile subsystem reliably handled authentication, notification delivery, Firestore synchronization, and analytics computation.

Test	Feature	Description	Condition	Expected Result	Result
T1	Authentication	Log in using valid and invalid credentials	Test user accounts in Firebase Auth	Valid logins succeed; invalid logins return appropriate error messages.	Pass
T2	Alert Retrieval	Confirm alerts appear in real time	Insert sample documents into alerts collection	New alerts populate Unacknowledged tab without refresh.	Pass
T3	Notifications	Receive alert while app is in the foreground	App open; send FCM test alert	Alert triggers a local notification; tapping opens correct detail screen.	Pass
T4	Notifications	Receive alert while app is backgrounded or closed	App minimized or killed	System notification delivered; tapping navigates to alert detail screen.	Pass
T5	Acknowledgment	Acknowledge alert and verify Firestore update	Existing alert in database	Fields acknowledged, acknowledgedBy, and acknowledgedAt update correctly.	Pass
T6	Analytics	Validate counts, room activity, and response time calculations	Alerts generated across multiple rooms	Analytics tab displays accurate totals, timestamps, and average response time.	Pass
T7	Logout	Test sign-out functionality	User authenticated	User redirected to login screen; main app inaccessible until re-authenticated.	Pass
T8	Notification Reliability	Send 200 alerts and verify delivery consistency	Batch of 200 FCM alerts sent to device	Device receives all 200 notifications with correct routing behavior (0 missed).	Pass
T9	Notification latency	Measure alert delivery time from Firestore to device	Alerts sent individually while timing with stopwatch	Each notification arrives within 5 seconds of dispatch.	Pass

Table 4: Mobile Subsystem Validation Test

5.4 Subsystem Conclusion

The mobile application subsystem met its objectives by providing a dependable way for users to receive, review, and acknowledge fall-detection alerts generated by the HARD system. Through its integration with Firebase Authentication, Cloud Firestore, and Firebase Cloud Messaging, the app supported secure access, real-time alert updates, and consistent notification delivery. Testing showed that alerts were displayed accurately, acknowledgements were recorded correctly, and the app remained synchronized with the backend at all times.

The subsystem also delivered basic analytics features, allowing users to view alert trends, room activity, and response times. These capabilities established a useful foundation for future development, including expanded analytics, UI refinements, and support for additional device platforms. Overall, the mobile subsystem functioned reliably and provided the necessary interface between the HARD detection hardware and end users for this phase of the project.